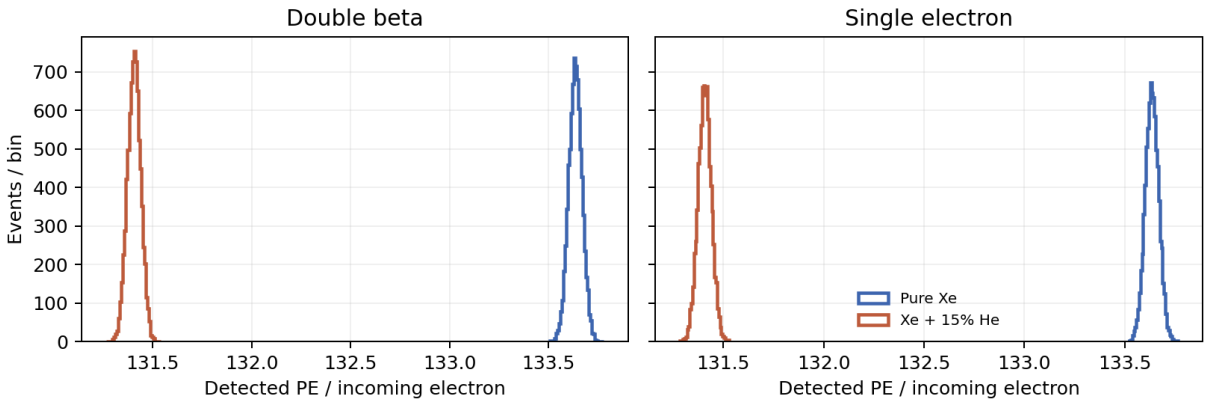
# Direct ion pad and electron SiPM readout

NAUSICAA analysis

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6 Julia processes sampled and collected 8,000 events from each MC sample in 55.97 s. Each retained sampled carrier was assigned directly to its physical pad or EL hole and longitudinal bucket. Pure-Xe and Xe–15% He electron arrivals were reduced to detected photoelectrons (PE) per SiPM and time bucket.

| Sample | Readout | Input/event | Signal/event | Signal/input | Buckets/event |
|---|---|---|---|---|---|
| bb0nu | ions | 112,143 | 112,143 | 1.00 | 1,584 |
| bb0nu | pure Xe $e^-$ | 112,106 | 14,981,373 | 133.64 | 1,005 |
| bb0nu | Xe–He $e^-$ | 112,129 | 14,734,402 | 131.41 | 396 |
| electrons | ions | 112,143 | 112,143 | 1.00 | 1,879 |
| electrons | pure Xe $e^-$ | 112,114 | 14,982,550 | 133.64 | 1,236 |
| electrons | Xe–He $e^-$ | 112,132 | 14,734,791 | 131.41 | 489 |




**Grid and timing inputs.** Ion pads have 2 mm pitch and 2 mm longitudinal buckets (20 ms at 0.1 m/s drift speed). Electron holes have 10 mm pitch and 5 mm diameter, with 5 mm EL depth. PE use 2 mm buckets (2 microseconds) at the configured electron drift speed. The ion and electron arrival orders are opposite. An independent uniform phase is used for each physical grid; the electron phase is shared by the two gas models of an event. The EL reduced field is 3 kV/cm/bar at 15 bar, with gate/anode potentials 0/22500 V in a relative voltage gauge.

**Optical model.** The fitted mean EL yields from the Xe–He study are 2332.4 and 2293.5 photons per electron for pure Xe and Xe–15% He. Photons are isotropic along the hole axis; VUV rays touching the walls are absorbed. The analytic direct-light acceptance is 0.190983 and the configured SiPM PDE is 30%. The fitted yield is held at its mean; PE counts are sampled with an exact binomial draw from the combined optical acceptance and PDE. PE are placed in the parent electron's arrival bucket; in-hole transit delay and electronics noise are outside this readout stage. The Xe–He EL fits were measured near 1.2 bar; their use at 15 bar applies reduced-field scaling.

**Validation.** The ion signal equals the retained input ion count in every event. Saved grid indices and time buckets account for each signal; pure-Xe and Xe–He readouts share a transverse grid phase per event. The finite diffusion window is the source-hit extent plus 3 diffusion widths on each side. The fractions outside that window in the two ion streams are 0.0171% and 0.0142%, respectively. The complete event-wise validation results accompany the HDF5 run summary.